

“Torsion pairs and 3-fold flops” (2025).

E-print available at [arXiv:2502.05146](https://arxiv.org/abs/2502.05146).

The paper classifies t-structures in the derived category of a 3-fold flopping contraction that have small cohomological spread with respect to the standard algebraic heart. This is a foundational step towards describing all t-structures and spherical objects in the derived category. Immediate applications of the results, to the classification of spherical modules, bricks, and semibricks are given.

In equivalent algebraic terms, the paper describes the complete lattice of torsion classes for an affine preprojective algebra. For this, new convex-geometric techniques are introduced which enable an arbitrary t-structure to be approximated by algebraic ones. The techniques have applications to other classes of finite-dimensional algebras.

“Simple complexes on a flopped curve” (2025).

E-print available at [arXiv:2508.06437](https://arxiv.org/abs/2508.06437).

The paper classifies, completely and without any caveats, all spherical objects and stability conditions on a single-curve flopping contraction. Consequently, long-standing conjectures about connectedness and contractibility of the stability manifold are shown to have affirmative answers in this context. In fact a stronger result is established, which classifies all objects satisfying a natural cohomological constraint. Algebraically, this is the first classification of tilting complexes on a g-affine algebra. This is proof of concept that the results of “*Torsion pairs and 3-fold flops*”, with appropriate geometric insight, can extend to global classification results.

“Kleinian orbifolds, cohomological Hall algebras, and Yangians” (2025),

joint with Francesco Sala *and* Olivier Schiffmann.

E-print available at [arXiv:2511.08576](https://arxiv.org/abs/2511.08576).

The paper computes cohomological Hall algebras for every Abelian category appearing as the heart of a stability condition on the minimal resolution of a Kleinian singularity. In particular CoHAs are computed for a new class of surfaces with non-commutative structure, namely those obtained as orbifold resolutions. The key technical tool is the approximation-based analysis of “*Torsion pairs and 3-fold flops*”, which allows for the natural t-structure on these surfaces to be approximated by well-understood algebraic ones.

Preprojective algebras and minimal surfaces are respectively the standard algebraic and geometric resolutions of a Kleinian singularity, and their cohomological Hall algebras are known to coincide with distinct ‘positive halves’ of the Maulik–Okounkov Yangian. Orbifold resolutions form a family of part-algebraic, part-geometric resolutions that interpolate between the two extremes. The computation of these cohomological Hall algebras recovers every positive half of the elliptic Lie algebra in the non-equivariant setting, and new, non-standard halves of Yangians in equivariant settings.